

Maedeh Agharazidermani

m.agharazi.d@gmail.com

Information Science researcher with postdoctoral experience leading human-subjects studies of mobile and wearable health technologies. Expertise in HCI, consumer health informatics, health-information behavior, mixed methods, IRB-regulated research, usability evaluation, and interdisciplinary collaboration. Current scholarship spans explainable and patient-facing AI, rehabilitation technology, and responsible use of health data.

Education

Ph.D. of Information, School of Information (iSchool), College of Communication and Information
Florida State University (FSU), Tallahassee, FL

Advised by Dr. Michelle Kazmer

Dissertation title: eHealth literacy, Information Overload, and Information Avoidance in the Era of the COVID-19

M.S. Sociocultural and International Development Education Studies, Department of Educational Leadership & Policy Studies, College of Education. **FSU**, Tallahassee, FL.

M.A. Multimedia (e-learning technologies), Multimedia University (MMU), Cyberjaya, Malaysia.

Thesis title: Perception of Microblogging especially Twitter users in learning new English vocabularies

B.S. Information System Engineering, MMU, Cyberjaya, Malaysia.

Thesis title: Map the Star

Research Experience

Collaborating Faculty, Curriculum Co-Design Research, HEX AI Lab, University of Pittsburgh, Department of Health Information Management —Spring 2025-present

- Engaged in a cross-institutional research collaboration in use of AI in patient engagement, AI explainability, and Federated learning method to enhance health data results and coauthoring scholarly outputs
- Co-development of online courses for clinicians about the use of AI in healthcare

Research collaboration, University of Florida (UF), Department of Physical Medicine and Rehabilitation—Summer 2024-fall2025

- Contributing to the development of AI-assisted technologies aimed at supporting post-stroke patients in home-based rehabilitation following hospital discharge.
- Engaged in ideation, literature review, and grant writing to support project advancement and funding acquisition.
- Collaborating within a multidisciplinary team to integrate clinical insights and user-centered design principles and instructional design concepts into the app's development.

Postdoctoral Associate, UF, College of Engineering—Spring 2022-Summer 2024—Under supervision of Dr. Boyer

- **Led a usability study on a smartwatch application for collaboration between runner dyads—Spring 2023-Summer 2024.**
 - Actively participated in meetings with developers, providing insights to enhance functionality and user experience and refining applications tailored for a variety of smartwatches.

- Ensured usability through rigorous testing of developed applications.
 - As Principal Investigator (PI) for the Collaborative Running Study, managed IRB documentation, RedCap documentation and consent forms, and the recruitment process both online and in-person.
 - Preparing and conducting survey questionnaires pre and post-study in the Qualtrics in addition to the pre and post-study interviews with participants.
 - Preliminary qualitative analysis of the collected data for a prospective thematic analysis paper.
- **Led a usability study on a phone application facilitating collaboration between cyclist dyads—Fall 2022-Spring 2023.**
 - Actively participated in meetings with developers, providing insights to enhance functionality and user experience.
 - Mentored two undergraduate students in application development.
 - Conducted multiple testing sessions and led the usability study to evaluate application usability.
 - Prepared Institutional Review Board (IRB) documentation, designed screening processes, and recruitment process.
 - Conducted interviews to gather participant feedback and analyzed the collected data which led to the authorship of a paper detailing study results.
- **Research assistant in Children’s heart rate study—Fall 2023**
 - Participated in data collection of a research testing a smartwatch application with children aged 9-12 to evaluate children’s heart rate comprehension and usability of the smartwatch application.
 - Assisted in data analysis and synthesized findings which led to writing a research article.

Research Assistant (volunteer), FSU, College of Psychology—Fall 2020-Fall 2021—Under supervision of Drs. Walter Boot and Zhe He

- Collaboration in qualitative data analysis and publishing an article on the motivational causes of participation of older adults in research projects. This was part of an NIH-funded project, Adherence Promotion with Person-centered Technology (APPT) to investigate promoting Adherence and Enhances the Early Detection and Treatment of Cognitive Decline. The project investigated AI-based reminder systems to help ensure long-term engagement with home-based cognitive assessment and cognitive training protocols.

Research experience, DIS, FSU, iSchool—Spring 2019—Under supervision of Dr. Chris Hinnant

- Conducting research on theoretical concepts including trust and data-driven decision-making approaches toward using big data in medium and small-size companies.
- Acquired expertise in crafting effective survey questions and developing questionnaires in Qualtrics, and implementing strategies to enhance response rates
- Initiated the exploration of fundamental concepts in Python programming.

Research Assistant, FSU, iSchool—Fall 2018-Summer 2019—Under supervision of Dr. Yolanda Rankin

- Contributed to qualitative research analysis for an FSU-funded project investigating the underrepresentation of African-American women in the computing pipeline using N-Vivo.

- Writing a paper examining the impact of family dynamics on the engagement of African-American women in computing based on the collected interviews.
- Provided mentorship to an undergraduate student, offering guidance in basic statistical data analysis, instruction in utilizing N-Vivo software and coding within it, and training in qualitative data analysis techniques.

Research Assistant, FSU, College of Education—Spring 2017—Under supervision of Dr. Helen Boyle

- Analysis of qualitative data collected from the Islamic schools in Ghana project led by Dr. Boyle, involving interviews with teachers, principals, parents, and religious leaders. Initial results were presented at the American Educational Research Association (AERA) conference.

Grant Writing Experience

Participated in grant writing—Under supervision of Drs. Boyer and Jake-Schoffman—Spring 2023–Spring 2024

Unfunded

“Fostering Parent-Child Physical Activity through Real-Time Collaboration and a Novel Smartwatch App: A Feasibility Pilot”

R21: NIH Exploratory/Developmental Research Grant Program (Parent R21 Clinical Trial Required), NIH Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD)

Submitted February 2023, Amount: \$275,000

- Participated in meetings, discussions, and brainstorming sessions with interdisciplinary team members to form the proposed project.
- Conducted literature reviews and synthesized relevant research findings to support the rationale and theoretical framework of the grant proposal.

“Moved2Learn: Empowering Mother-Daughter Dyads to Learn AI in Support of their Physical Activity Goals”

Advancing Informal STEM Learning (ASIL), NSF Division of Research on Learning

Submitted January 2023, Amount: \$1,491,229

- Participated in meetings, discussions, and brainstorming sessions with interdisciplinary team members to form the proposed project.
- Conducted literature reviews and synthesized relevant research findings to support the rationale and theoretical framework of the grant proposal.

“DRPD-ROSF2024: Harnessing the Power of Artificial Intelligence to Promote Physical Activity and Community Building among College Students”

UF Research Opportunity Seed Grant, University of Florida

Submitted February 2024, Amount: \$100,000

- Participated in meetings, discussions, and brainstorming sessions with interdisciplinary team members to form the proposed project.
- Conducted literature reviews and synthesized relevant research findings to support the rationale and theoretical framework of the grant proposal.

“Fostering Collaborative Parent-Child Physical Activity through a Novel AI-Powered App: A Feasibility Pilot”

R21: NIH Exploratory/Developmental Research Grant Program (Parent R21 Clinical Trial Required), NIH Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD)

Submitted February 2024, Amount: \$275,000

- Collaborated closely with the PIs to refine project ideas, develop research aims, and articulate the significance and innovation of the proposed project.
- Contributed to the formulation of research questions and strategies for integrating AI-supported mobile app technology into the intervention design.
- Participated in meetings, discussions, and brainstorming sessions with interdisciplinary team members to form the proposed project.
- Conducted literature reviews and synthesized relevant research findings to support the rationale and theoretical framework of the grant proposal.

Teaching experience

Adjunct professor, University of Illinois Urbana-Champaign, iSchool—September 2025-present

Adjunct professor, Florida State University, ischool—December 2024-present

- Health Information Systems and Management (LIS 5788), Spring 2025, 2026

Lead Instructor, FSU, ischool, Fall 2020

- Information Ethics in 21st Century (LIS 2144), Fall 2020

Teaching Assistant, online and face-to-face courses, FSU, iSchool—Fall 2015-Summer 2021

- Information Architecture (LIS5786)
- Design and Production of Networked Multimedia (LIS5362)
- Digital Media (LIS5313)
- Digital storytelling (LIS2527)
- Management of Information Organizations (LIS5408)
- Assessing Information needs (LIS5203)
- Information organization (LIS5703)
- Perspective in information technology (LIS4708)
- Empowering Health Consumers in the eHealth Era (IFS 3037)

Publications

Refereed Proceedings and Journal Articles

1. Rezvani, F., Bogden, E., Mathew, J., **Agharazodermanni**, M., Farahani, S., Pourhasani, P., Bruno, E., Kann, M., Horne, T., Myers, N., Davis, J., Gupta, P., Plate, J., Tafti, A., Ghazaleh, H. (2026). *Trustworthy federated learning for postoperative respiratory risk prediction following total joint arthroplasty*. **Manuscript in preparation.**
2. Pourhasani, P., Farahani, S., Rezvani, F., Bogden, E., Mathew, J., Bruno, E., Kann, M., Horne, T., Myers, N., Davis, J., Gupta, P., Plate, J., Ghazaleh, H., **Agharazodermanni**, M., Tafti, A. (2026). *Federated Learning for Identifying Demographic Disparities in Cancer Survivorship Across Multi-Institutional Cohorts*. **Manuscript in preparation.**

3. Bogden, E., Rezvani, F., Bruno, E., Poorhasani P., Farahani, S., **Agharazidermani, M.**, Mathew, J., Urish, K., Plate, J., Ghazaleh, H., Tafti, A. (2026) *Explainable Federated Learning to Predict Cardiac Complications Following Total Hip and Knee Arthroplasty*. *AMIA*
4. Sanogo, M., Gao, F., Littlefield, N.,..., **Agharazidermani, M.**,... *et al.* KneeXNet-2.5D: a clinically-oriented and explainable deep learning framework for MRI-based knee cartilage and meniscus segmentation. *npj Health Syst.* 3, 18 (2026).
<https://doi-org.proxy2.library.illinois.edu/10.1038/s44401-026-00072-5>
5. Sharififar, S., **Agharazidermani, M.**, Azari, H., Yuzefovich Khanna, A., Adkins, L., Babb, C., Gonzalez, N., & Rashidi, P. (submitted). *Technology-driven assessment of gait deviations in stroke rehabilitation: A systematic review of methods and outcomes*. PROSPERO systematic review registration.
6. Lu, L., Jake-Schoffman, D. E., Lavoie, H. A., **Agharazidermani, M.**, & Boyer, K. E. (2025). Preadolescent Children Using Real-Time Heart Rate During Moderate to Vigorous Physical Activity: A Feasibility Study. *JMIR Human Factors*, 12(1), e58715.
7. **Agharazidermani, M.**, Lu, L., & Boyer, K. E. (2023, September). Exploring Real-Time Collaborative Heart Rate Displays for Cycling Partners. In *Proceedings of the 25th International Conference on Mobile Human-Computer Interaction* (pp. 1-7). Athens, Greece.
8. Carr, D., Tian, S., He, Z., Chakroborty, S., Dieciuc, M., Gray, N., **Agharazidermani, M.**, Lustria, M., Dilanchian, A., Zhang, S., Charness, N., Terracciano, A., Boot, W. (2022). Motivation to Engage in Aging Research: Are There Typologies and Predictors? *The Gerontologist*.
9. Rankin, Y., **Agharazidermani, M.**, Thomas, J. (2020). *The Role of Familial Influences in African American Women's Persistence in Computing*. In *Research on Equity & Sustained Participation in Engineering, Computing, & Technology (RESPECT) Proceedings*. March 2020 Portland, OR.
10. **Agharazidermani, M.**, Song, H., & Rahimi, S. (2011). *Microblogging as an educational tool to advance learning: case studies and recent reports*. In *EDULEARN11 Proceedings* (pp. 6181-6190). IATED. July 2011, Barcelona, Spain.
11. Rahimi, S., Song H., & **Agharazidermani M.** (2011). *Perception and experiences of undergraduate students on using second life as a learning tool*. In *EDULEARN11 Proceedings* (pp. 6181-6190). IATED. July 2011, Barcelona, Spain.

Referenced Presentations at Conferences

1. Sharififar, S., **Agharazidermani, M.**, Zadov, Y., Vincent, H.K., & Rashidi, P., (2025). *Admission Self-Care and Mobility Scores Predict Length of Stay and Discharge Outcomes in Stroke Rehabilitation*. American Academy of Physical Medicine and Rehabilitation (AAPM&R). Salt Lake City, Utah.
2. **Agharazidermani, M.**, Menard, W., Lavoie, H., Lu, L., Jake-Schoffman, D. E., & Boyer, K. E. (2024). Cycling is a Collaborative Sport and Cycling HCI Needs More Collaboration Research. *CyclingHCI*. CHI 2024, Honolulu, HI, USA.
3. **Agharazidermani, M.**, (2023) *Facilitating collaborative physical activity through real-time information sharing*. Postdoctoral Seminars. Gainesville, FL.
4. Carr, D. C., He, Z., Lustria, M. L. A., Tian, S., **Agharazidermani, M.**, Boot, W. R. (2021, November 10-13). *Factors That Motivate Older Adults to Participate in Research: Typologies and Implications* [Virtual Presentation]. Gerontological Society of America, Virtual Conference. (International)

5. Muhamad, J.S., Huse, L.K., Wibowo, P., Tian, M., Gunaydin, F., Wendorf Muhamad, J., Merle, P., & **Agharazidermani**, M. (2020). *Information Sharing for Collective Action: Exploring Retweets during the COVID-19 Pandemic*. Paper submitted to the 106th National Communication Association conference. Indianapolis, IN.
6. Huse, L.K., Muhamad, J.S., Wendorf Muhamad, J., Wibowo, M., Tian, M., Merle, P. Gunaydin, F., & **Agharazidermani**, M. (2020). *Baby Yoda Recommends Washing Hands": A Content Analysis of Memes on COVID-19*. Paper submitted to the 106th National Communication Association conference. Indianapolis, IN.
7. Merle, P., Wendorf Muhamad, J., Huse, L.K., Wibowo, P., Gunaydin, F., Muhamad, J.S., & **Agharazidermani**, M. (2020). *Who framed COVID-19 and how: Examining the local and national communication patterns on the pandemic*. Paper submitted to the 103rd Annual Association for Education in Journalism and Mass Communication conference. San Francisco, California.
8. Muhamad, J.S., Wendorf Muhamad, J., Tian, M., Gunaydin, F., Merle, P., Huse, L.K., Wibowo, P., & **Agharazidermani**, M. (2020). *"Chinese Virus" as Anchor for Engaging with COVID-19 Information: Anchor Bias leading to Racism and Xenophobia*. Paper submitted to ALISE 2020 annual conference. Pittsburgh, PA.
9. Wibowo, P., Wendorf Muhamad, J., Muhamad, J.S., & **Agharazidermani**, M. Gunaydin, F., Merle, P., Huse, L.K., Tian, M. (2020). *Analysis of Public Perceptions of Multiple Community Issues through Social Media Mining during a Pandemic*. Paper submitted to ALISE 2020 Annual conference. Pittsburgh, PA.
10. Boyle, H. N., & **Agharazidermani**, M. (2017). *Blending Islamic and Western Traditions of Education: Integrated Islamic Schools in Ghana*. Paper Presented at American Educational Research Association, San Antonio, TX.

Awards and Fellowships

- Best paper award for Late-breaking papers at Mobile HCI23 conference, 2023.
- COGS travel grant for attending AMIA21 conference, 2021.
- School of Information's Outstanding Teaching Assistant Award, Spring 2019.
- William A. Kerr Fellowship award, \$6,000, FSU, 2016.

Guest Lectures

- Information Behavior—graduate course FSU iSchool (LIS 5203)—Information Consumption, Information Overload, and Information Avoidance in the Era of the COVID-19

Administrative/technical Experiences:

Graduate Assistant, FSU, iSchool—Fall 2019

- Preparing enrollment reports and providing recommendations to improve enrollment rates for the master of Library Science at FSU.
- Participated in data analysis for Quality Enhancement Reviews (QER) focusing on student experiences. Additionally, collaborated within FSU's iSchool to create reports, develop data visualizations, and facilitate requirements for ALA reviews conducted every seven years.

- Assisted the admission office to collect data by conducting phone surveys with alumni and current students to investigate their experience in the Master of Library Science program.

Administration Assistant, FSU, iSchool—Summer 2017

- Assisted the director with the tasks related to student enrollment and document preparation documents for the admissions committee to facilitate decision-making regarding prospective students' admission.

Graduate Assistant, FSU, College of Education—Spring 2016, Under supervision of Dr. Peter Easton, chief editor of CIES journal

- Bibliography and database management of the Comparative and International Education Society (CIES) journal.

Graduate Assistant, FSU, College of Education—Spring 2016, Under supervision of Dr. Boyle

- Assisted in instruction development and redesign of the course content to enhance student engagement and alignment with learning outcomes.
- Developed the syllabus and defined learning objectives in collaboration with the course instructor.
- Identified core content areas and designed in-class activities to support diverse learning needs and promote inclusive teaching practices

Instructional Designer, Cosmopoint, Malaysia —December 2010-December 2011

- Cosmopoint was a part of a Malaysian company that supported Kuala Lumpur Metropolitan University (KLUM). In that role, I was responsible for helping instructors create their courses based on the instructional design principles, designing and developing multimedia content, and I was involved with conducting workshops for instructors to teach them how to use PowerPoints effectively in their classrooms.

IT Professional Technician, Tehran, Iran—March 2007- June 2007

- I was working in the Iranian National Science Foundation (INSF) IT helpdesk. Technical troubleshooting, conducting workshops on how to use the new automation database system, and using the new system were part of my organization's responsibility.

Internship, Kuala Lumpur, Malaysia—June 2007-October 2007

- IBTech SDN BHD was a Malaysian tech company, and as an intern, I was involved with the R&D team to facilitate designing a knowledge management system and search engines

Audio/Visual Projects

- **Producer and Co-Creator, *T-Pod*** — A podcast dedicated to exploring questions about learning and education. In each episode, two hosts respond to audience questions through evidence-based discussions on topics related to learning sciences and cognitive processes, aiming to make academic insights accessible to a broader audience.
- Rahimi, S., **Agharazidermani, M.** (2017, July). Interview with John Keller - The beautiful mind behind the ARCS model of motivation [original video]. Tallahassee, FL. See the interview here: <https://youtu.be/VSEA0tX8e04>

Service

- AMIA reviewer
- AMIA 2021 Annual Symposium Student Volunteer Program
- Mentoring and teaching an undergraduate student to work in Detour lab on data analysis on the project of underrepresentation of African American women in computing.

- Doctoral Students' representative in the Student Experience Committee at school of information (2019-2020)

Membership in Professional Organizations

- Association of Computing Machinery (ACM)
- Association for Library and Information Science Education (ALISE)
- American Medical Informatics Association (AMIA)

Certificates

- AI Fundamentals—Certificate of completion by Data Camp—2021
- Social network analysis using R—certificate of completion by Lynda.com—June 2020
- Learning Python—certificate of completion by Lynda.com —March 2019
- Program Evaluation certificate by College of Education in Florida State University— 2017
- A two-day workshop including contents like Needs Assessment, design a learning system, implement a system, and evaluate its outcomes—July 2016
- Developing ASP.NET Web Applications and XML Web Services with Microsoft Visual C# .NET and Microsoft Visual Studio.NET certificate given by Microsoft—2006
- Installing, Configuring, and Administering Microsoft SQL Server 2000 Enterprise Edition certificate issued by Microsoft—2006

Software/Applications

N-Vivo, R, Mplus, Weka, SPSS, HTML, Python, Microsoft Office, Adobe Photoshop, Premier, Dreamweaver, Articulate Storyline.